23 (a) Show that the equation $2x^3 + 4x = 3$ has a solution between 0 and 1

Let $F(x) = 2x^3 + 4x - 3$.

$F(0)=-3$; $F(1)=2+4-3=3$.

1 mark
Opposite signs

F is continuous; the sign change proves a root in(0,1).

1 mark
Continuous sign change

(2)

(b) Show that $2x^3 + 4x = 3$ can be rearranged to give $x = \frac{3}{4} - \frac{x^3}{2}$

$4x = 3 - 2x^3$

Divide by4: $x = 3/4 - x^3/2$.

1 mark
Rearrangement shown

(1)

(c) Starting with $x_0 = 0$, use the iteration formula $x_{n+1} = \frac{3}{4} - \frac{x_n^3}{2}$ three times to find an estimate for the solution to $2x^3 + 4x = 3$

Start $x_0=0$; next = 0.75 - (previous)$^3/2$.

$x_1=0.75 - 0^3/2 = 0.75$

1 mark
First iteration

$x_2=0.75 - 0.75^3/2 = 0.5390625$

$x_3=0.75 - 0.5390625^3/2 = 0.671677351...$

1 mark
Two more iterations

1 mark
Three-step estimate0.671677351...

(3)

(Total for Question 23 is 6 marks)